UNE APPROCHE BASÉE INCERTITUDE POUR TRANSPORTER UN SIGNAL D'UN GRAPHE À UN AUTRE

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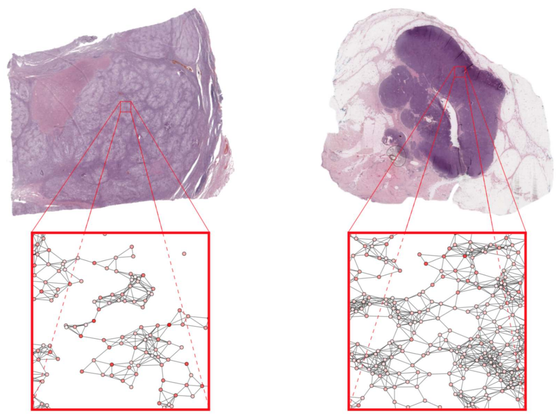
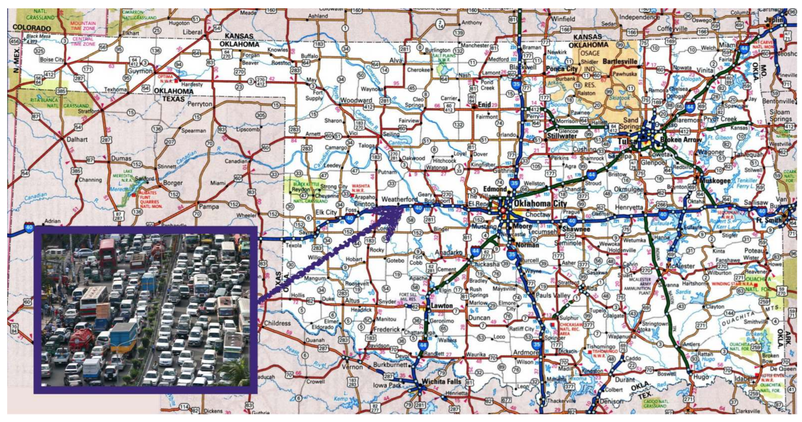
Motivation



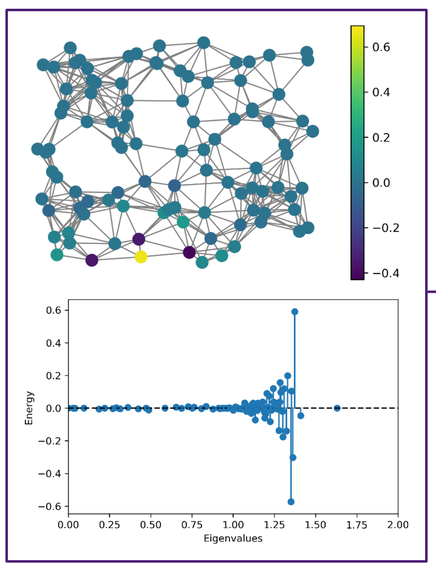
← La répartition des cellules est-elle la même ?
Trouver des patients similaires et suggérer des traitements

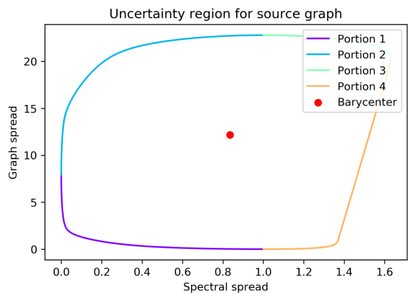
Cette situation de trafic a-t-elle été observée dans une autre ville ? →
Anticiper un possible bouchon en régulant le trafic

Peut-on transporter un signal x_1 d'un graphe G_1 à un graphe G_2 ?



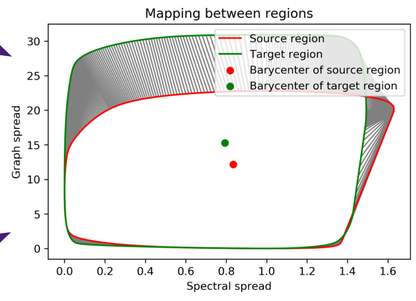
Méthodologie

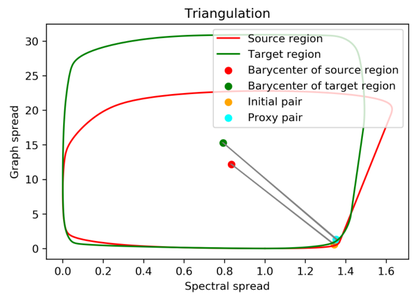


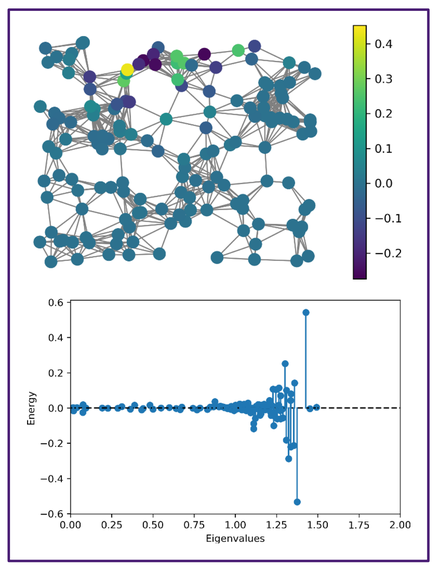


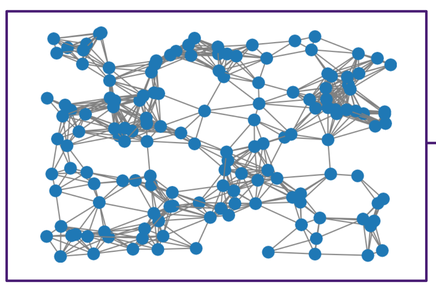
Graph spread [1] : $\Delta_{\mathcal{G},v_c}(\mathbf{x}) = \frac{1}{\|\mathbf{x}\|_2} \sum_{v \in \mathcal{V}} d_{\mathcal{G}}(v_c, v)^2 \, \mathbf{x}[v]^2$

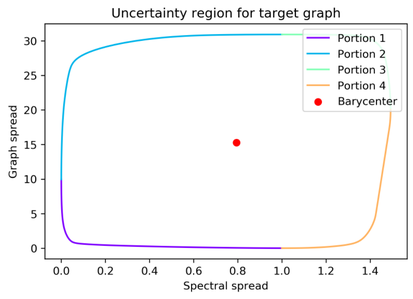
Spectral spread [1] : $\widehat{\Delta}_{\mathcal{G}}(\mathbf{x}) = \frac{1}{\|\mathbf{x}\|_2} \sum_{\lambda \in diag(\mathbf{\Lambda})} \lambda^2 \, \widehat{\mathbf{x}}[\lambda]^2$











$$\mathbf{x}'_1 \in \underset{\mathbf{x} \in \mathbb{R}^{N_2}}{\text{Argmin}} \; H(\mathbf{x}) \quad t.q. \quad \begin{cases} \|\mathbf{x}\|_2 = 1 \, , \\ \left(\widehat{\Delta}_{\mathcal{G}}(\mathbf{x}), \Delta_{\mathcal{G},v_{c_2}}(\mathbf{x})\right) = p'_1 \end{cases}$$

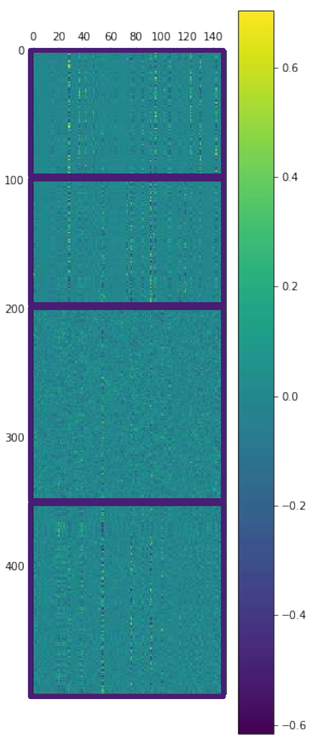
Résultats

Chebyshev [2] (K=5) sur $G_1 \rightarrow G_2$

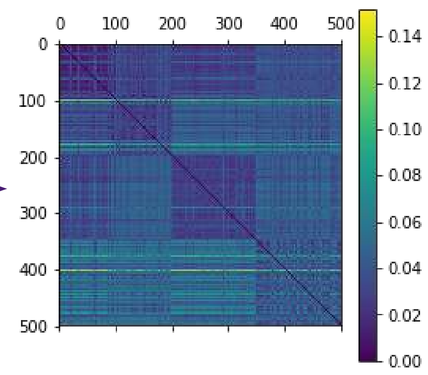
Chebyshev [2] (K=20) sur $G_1 \rightarrow G_2$

Chebyshev [2] (K=5) sur G_2

Chebyshev [2] (K=20) sur G_2

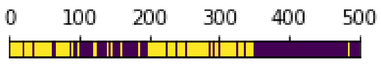


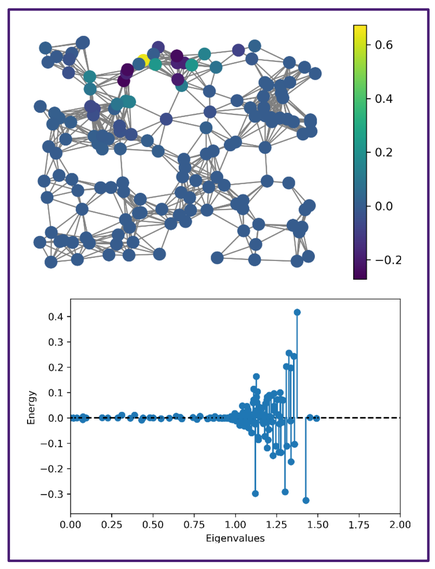
Norme l_2 de la différence



Partitionnement hiérarchique [3]

	$K=5$	$K=20$
Rappel	0.848	0.908
Précision	0.902	0.857





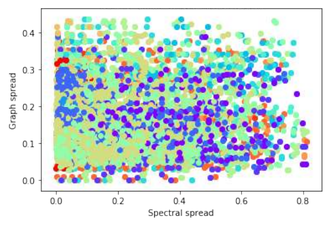
Contrôle – Chebyshev sur G_2

Conclusions & extensions

- Premiers résultats encourageants
- Abstraction du choix de v_c →
- Etude de nouvelles classes de signaux et de datasets réels

$$\Delta_{\mathcal{G}}(\mathbf{x}) = \min_{v_c \in \mathcal{V}} \frac{1}{\|\mathbf{x}\|^2} \sum_{v \in \mathcal{V}} d_{\mathcal{G}}(v_c, v)^2 \, \mathbf{x}[v]^2 = \min_{v_c \in \mathcal{V}} \frac{1}{\|\mathbf{x}\|^2} \mathbf{x}^\top \mathbf{P}^2_{\mathcal{G},v_c} \mathbf{x}$$

$$\widehat{\Delta}_{\mathcal{G}}(\mathbf{x}) = \min_{\lambda_c \in diag(\mathbf{\Lambda})} \frac{1}{\|\mathbf{x}\|^2} \sum_{\lambda \in diag(\mathbf{\Lambda})} (\lambda_c - \lambda)^2 \widehat{\mathbf{x}}[\lambda]^2 = \min_{\lambda_c \in diag(\mathbf{\Lambda})} \frac{1}{\|\mathbf{x}\|^2} \mathbf{x}^\top (\mathbf{L} - \lambda_c \mathbf{I}_N)^2 \mathbf{x}$$



Références

[1] A. Agaskar and Y. M. Lu – A spectral graph uncertainty principle – IEEE Transactions on Information Theory (2013).

[2] D. I. Shuman, P. Vandergheynst and P. Frossard – Chebyshev polynomial approximation for distributed signal processing – DCOSS (2011).

[3] J. H. Ward Jr. – Hierarchical grouping to optimize an objective function – Journal of the American statistical association (1963).

Codes : https://github.com/BastienPasdeloup/gretsi2019